Model Inspired Autoencoder for Unsupervised Hyperspectral Image Super-Resolution

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***Abstract*— This article focuses on hyperspectral image (HSI) super-resolution that aims to fuse a low-spatial-resolution HSI and a high-spatial-resolution multispectral image to form a high-spatial-resolution HSI (HR-HSI). Existing deep learning-based approaches are mostly supervised that rely on a large number of labeled training samples, which is unrealistic. The commonly used model-based approaches are unsupervised and flexible but rely on handcrafted priors. Inspired by the specific properties of model, we make the first attempt to design a model-inspired deep network for HSI super-resolution in an unsupervised manner. This approach consists of an implicit autoencoder network built on the target HR-HSI that treats each pixel as an individual sample. The nonnegative matrix factorization (NMF) of the target HR-HSI is integrated into the autoencoder network, where the two NMF parts, spectral and spatial matrices, are treated as decoder parameters and hidden outputs, respectively. In the encoding stage, we present a pixelwise fusion model to estimate hidden outputs directly and then reformulate and unfold the model's algorithm to form the encoder network. With the specific architecture, the proposed network is similar to a manifold prior-based model and can be trained patch by patch rather than the entire images. Moreover, we propose an additional unsupervised network to estimate the point spread function and spectral response function. Experimental results conducted on both synthetic and real datasets demonstrate the effectiveness of the proposed approach.**

***Index Terms*— Autoencoder, hyperspectral image (HSI), nonnegative matrix factorization (NMF), super-resolution, unfolding.**

I. Introduction

HYPERSPECTRAL image (HSI) is a kind of 3-D image taken at different spectral bands, with its spectral range covering hundreds of contiguous and narrow bands that span the visible to infrared spectrum. The high spectral resolution of HSIs enables accurate identification of materials and thereby promotes various applications, such as military, agriculture, and mineralogy. Due to the limited incident energy, there is always a tradeoff between spectral resolution, spatial resolution, and signal-to-noise ratio of images when designing the imaging sensors [1]–[6]. Thus, the spatial resolution of HSIs is usually sacrificed, which impedes the subsequent tasks. Conversely, conventional multispectral images (MSIs) at much lower spectral resolution can be acquired with higher spatial resolution. An economical HSI super-resolution solution is to instead record a low-spatial-resolution HSI (LR-HSI) and a high-spatial-resolution MSI (HR-MSI) and to fuse them into a target high-spatial-resolution HSI (HR-HSI) [2], [3], [5]. The HSI super-resolution solution breaks the limitations of the imaging sensors and has shown a good performance in practice.

HSI super-resolution that fuses an LR-HSI with an HR-MSI has attracted great attention [2], [3], [5]. This fusion problem arises from the pansharpening problem that fuses a low-spatial-resolution MSI or HSI with a high-spatial-resolution panchromatic image [1], [4], [6]. Generally, the conventional approaches proposed for the pansharpening problem can be extended to solve HSI super-resolution, but the fusion process of HSI super-resolution is relatively more complicated than that of pansharpening due to the rich spectral information. Related fusion approaches can be roughly divided into four categories: component substitution [7], multiresolution analysis [8], model-based approaches, and deep learning-based approaches. Among these categories, the research of model-based approaches is the most classic one, and deep learning-based approaches have been the most active one recently.

Model-based approaches consider building optimization models to obtain the target image. Given two observed images, they design fidelity terms and exploit spectral/spatial priors to enforce the desired result. Some approaches treat the target image as a variable and recover the target image entirely, such as group spectral embedding [9], clustering manifold structure [10], nonlocal patch tensor sparse representation [11], and structured sparse low-rank representation [12]. Most approaches consider separating the target image into parts and regenerating it via the recovered parts. There are many decomposition strategies by making assumptions about the target image. Examples are that it lives in a low-dimensional subspace and the subspace-based models are solved by exploiting prior knowledge, such as piecewise smooth [13], dictionary learning [14], tensor multirank [15], low tensor-train

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rank [16], and truncated matrix decomposition [17]; or that it can be represented linearly by pure spectral signatures and the endmember and abundance matrices are recovered simultaneously [18]–[21]; or that it can be sparsely represented by an overcomplete spectral dictionary and different priors are used to obtain the spectral dictionary and coefficients [22]–[26] or by approaches that separate the target image by tensor decomposition and update each component iteratively [27]–[31]. Moreover, there are some approaches that build models to estimate the point spread function (PSF) and spectral response function (SRF) [13], [32]. The entire process of model-based approaches is unsupervised. Although these models are flexible and their theory is relatively complete, they rely on handcrafted priors and there are many empirical parameters to tune.

Deep learning-based approaches are data-driven. They build deep neural networks to solve the related fusion problems and produce the target image by feeding observed images into the network. Some approaches enhance the ability to fuse images in the network structures, such as 3-D convolutional neural networks (CNNs) [33], residual networks [34], multiscale structures [35], pyramid networks [36], attention networks [37], [38], cross-mode information [39], dense networks [40], [41], and adversarial network [42]–[44]. Some approaches use detailed information from high-spatial-resolution conventional images to improve the performance [45]–[48]. Inspired by the specific properties of model, some form a hybrid of model- and deep learning-based approaches [49]–[51], and some use the deep unfolding technique to ease the construction of networks [52]–[55]. These approaches have shown good performance in exploiting the relationship between the observed and target images. However, they are mostly supervised that require plenty of labeled samples to train the networks, which limits their applications in many scenarios.

There are some deep learning-based approaches developed for HSI super-resolution that are performed in an unsupervised manner. For instance, Dian *et al.* [56] introduced a CNN denoiser to regularize the fusion model. Zhang *et al.* [57] integrated the deep image prior into the fusion model and thereby presented a unified unsupervised network for HSI super-resolution. Qu *et al.* [58] exploited an unsupervised approach composed of two autoencoder networks, which are coupled through a shared decoder. Wang *et al.* [59] proposed a variational probabilistic autoencoder framework implemented by CNNs for HSI super-resolution. Yao *et al.* [60] proposed a two-stream convolutional autoencoder framework inspired by coupled spectral unmixing (CSU) and introduced a cross-attention module to improve the performance. Uezato *et al.* [61] designed a network composed of an encoder–decoder network and a deep decoder network. Zheng *et al.* [62] proposed a network consisting of three coupled autoencoder networks, inspired by CSU, where the three autoencoder networks are coupled through two convolutional layers. Most approaches are built on the autoencoder architecture. Similar to model-based approaches, the construction of networks relies too much on human experience.

Inspired by the specific properties of model, we consider constructing an unsupervised network by referencing some models and propose a model-inspired autoencoder (MIAE) for unsupervised HSI super-resolution. Specifically, we perform nonnegative matrix factorization (NMF) on the target HR-HSI to maintain its intrinsic structure and thereby propose an implicit autoencoder network for HR-HSI by integrating its NMF model. In the autoencoder network, each hyperspectral pixel is treated as an individual sample, and the two NMF parts of the target HR-HSI, i.e., spectral and spatial matrices, are treated as decoder parameters and hidden outputs, respectively. Since the inputs of the autoencoder network are unknown, we take the two observed images as inputs and present a pixelwise fusion model to estimate each hidden output vector directly. The pixelwise fusion model is solved by the gradient descent algorithm, and the algorithm is reformulated and unfolded to form the encoder network. The loss function is just built on the mechanism of spectral and spatial degradations, and an additional blind estimation network is proposed to estimate the PSF and SRF. Compared with the existing HSI super-resolution approaches, some of the innovative characteristics of MIAE are highlighted as follows.

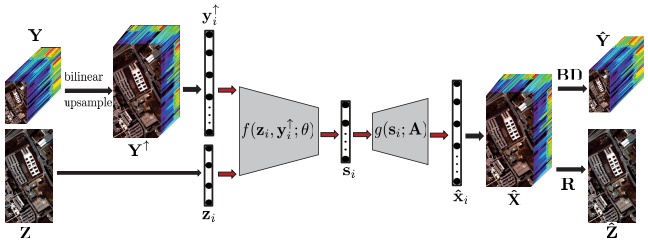


Fig. 1. Overall architecture of MIAE.

1) MIAE is an unsupervised deep learning-based approach that involves only one implicit autoencoder. The autoencoder network treats each pixel as an individual sample, and thus, the proposed network can be treated as a kind of manifold prior-based model and can be trained patch by patch to accelerate the training process.
2) MIAE is constructed by referencing models, and thus, the construction of the network is relatively concise. The NMF of the target HR-HSI is integrated into the autoencoder, and the encoder network is inspired by the pixelwise fusion model.
3) An additional unsupervised network is proposed to estimate the PSF and SRF from the two observed images directly.

The remainder of this article is organized as follows. Section II proposes the proposed MIAE and its relationship to the model-based approaches, as well as the blind estimation network. In Section III, the effectiveness of MIAE is demonstrated through experiments on three synthetic datasets and one real dataset. Section IV provides the concluding remarks.

II. Proposed Approach

Fig. 1 shows the overall architecture of MIAE. The details of the proposed network are described as follows.

A. NMF-Inspired Autoencoder

NMF is a useful dimension reduction method [63]. It can capture the intrinsic structure of the data and represent the

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data in a sparse manner. The properties of NMF indicate that it can facilitate the inference process of super-resolution if we perform NMF on the target HR-HSI. Let us represent the target HR-HSI as a matrix $\hat{\mathbf{X}} \in \mathbb{R}^{N_B \times N_H N_W}$, where N_B denotes the spectral band and N_H and N_W denote the spatial height and width, respectively. NMF aims to factor $\hat{\mathbf{X}}$ into two rank-J ($J < \min\{N_B, N_H N_W\}$) nonnegative matrices, i.e.,

$$\hat{\mathbf{X}} \approx \mathbf{AS}, \tag{1}$$

where the spectral matrix $\mathbf{A} \in \mathbb{R}^{N_B \times J} \succeq 0$ and the spatial matrix $\mathbf{S} \in \mathbb{R}^{J \times N_H N_W} \succeq 0$ with $\succeq$ being a componentwise inequality.

NMF can be integrated into an autoencoder network [64]–[66]. Equation (1) can be rewritten as

$$\hat{\mathbf{x}}_\mathbf{i} \approx \mathbf{As}_i \quad \forall i \tag{2}$$

where $i = 1, 2, \ldots, N_H N_W$ and $\hat{\mathbf{x}}_\mathbf{i} \in \mathbb{R}^{N_B}$ and $\mathbf{s}_i \in \mathbb{R}^J$ are the column vectors of $\hat{\mathbf{X}}$ and $\mathbf{S}$, respectively. Let $\hat{\mathbf{x}}_i$ represent the reconstructed vector and $\mathbf{s}_i$ represent the hidden output vector, and we can construct the following autoencoder network:

$$\mathbf{x}_i \xrightarrow{f(\cdot)} \mathbf{s}_i \xrightarrow{g(\cdot)} \hat{\mathbf{x}}_i \quad \forall i \tag{3}$$

where the input data $\mathbf{x}_i \in \mathbb{R}^{N_B}$ are the column vector of the observed HR-HSI $\mathbf{X} \in \mathbb{R}^{N_B \times N_H N_W}$. The network (3) consists of two networks $f(\cdot)$ and $g(\cdot)$. $f(\cdot)$ (described in Section II-B) is the proposed model-inspired encoder network with $\mathbf{s}_i = f(\mathbf{x}_i; \theta)$, where θ denotes all trainable parameters involved in the network. In hyperspectral unmixing [64]–[66], sum-to-one constraint is added to enforce the hidden vector, i.e., $\mathbf{1}_J^T \mathbf{s}_i = 1$ with $\mathbf{1}_J \in \mathbb{R}^J$ being a vector of all 1s. We do not intend to finish the two tasks of fusion and unmixing at once and only use the nonnegative constraint. Specifically, $\mathbf{s}_i$ is enforced using $\mathcal{C}_0^1(\mathbf{s}_i)$ when designing the network, where $\mathcal{C}_0^1(\cdot)$ is a clamp function that forces all elements of the input vector/matrix into the range [0, 1].

The decoder network $g(\cdot)$ derived from (2) is the reverse of the NMF process. It can be expressed as $\hat{\mathbf{x}}_i = g(\mathbf{s}_i; \mathbf{A}) = \mathbf{As}_i$, where the spectral matrix $\mathbf{A}$ is treated as the trainable weight matrix. When designing the network, the weight matrix $\mathbf{A}$ is enforced using $\mathcal{C}_0^1(\mathbf{A})$ and implemented by applying the clamp function every iteration. To restrict the scale of outputs, the clamp function is applied to $\hat{\mathbf{x}}_i$, i.e., $\hat{\mathbf{x}}_i = \mathcal{C}_0^1(\mathbf{As}_i)$.

If the input data $\mathbf{X}$ are given, one can build a loss function for the autoencoder network (3) as follows:

$$\mathcal{L} = \|\mathbf{X} - \hat{\mathbf{X}}\|_{1,1} \tag{4}$$

where $\|\cdot\|_{1,1}$ represents the absolute sum of all the matrix elements. Then, one can train the network (3) by feeding the $N_H N_W$ hyperspectral pixels.

B. Pixelwise Fusion Model-Inspired Encoder Network

In HSI super-resolution, the input data $\mathbf{X} \in \mathbb{R}^{N_B \times N_H N_W}$ are not given. One cannot train the network (3) directly. Instead, we have two observed (i.e., degenerated) images of $\mathbf{X}$, an LR-HSI $\mathbf{Y} \in \mathbb{R}^{N_B \times N_h N_w}$ and an HR-MSI $\mathbf{Z} \in \mathbb{R}^{N_b \times N_H N_W}$, where $N_b < N_B$ is the multispectral band and N_h and N_w are the spatial sizes. We assume that $N_H = rN_h$ and $N_W = rN_w$ with $r > 1$ being the resolution ratio. The observations $\mathbf{Y}$ and $\mathbf{Z}$ can be modeled as spatially degraded and spectrally degraded versions of $\mathbf{X}$. Specifically, these two degeneration processes can be written as

$$\mathbf{Y} \approx \mathbf{XBD} \tag{5}$$

$$\mathbf{Z} \approx \mathbf{RX} \tag{6}$$

where the PSF $\mathbf{B} \in \mathbb{R}^{N_H N_W \times N_H N_W}$ is the spatial blur, $\mathbf{D} \in \mathbb{R}^{N_H N_W \times N_h N_w}$ is the spatial downsampling, and $\mathbf{R} \in \mathbb{R}^{N_b \times N_B}$ is the SRF of multispectral sensor.

In (3), the network needs to be trained by feeding the input data pixel by pixel. It is the key to the success of autoencoder. For the degeneration processes, (6) can be rewritten as a pixelwise formulation, i.e., $\mathbf{z}_i \approx \mathbf{Rx}_i$ with $\mathbf{z}_i \in \mathbb{R}^{N_b}$ being the column vector of $\mathbf{Z}$, whereas (5) cannot because of the coupling matrices $\mathbf{B}$ and $\mathbf{D}$. We consider resizing $\mathbf{Y}$ to the same size as $\mathbf{X}$ using bilinear interpolation, in order to approximate $\mathbf{X}$ pixel by pixel at the spectral level. $\mathbf{x}_i$ can be obtained by solving

$$\min_{\mathbf{x}_i} \frac{1}{2}\|\mathbf{z}_i - \mathbf{Rx}_i\|_2^2 + \frac{\lambda}{2}\left\|\mathbf{y}_i^{\uparrow} - \mathbf{x}_i\right\|_2^2 \quad \forall i \tag{7}$$

where $\lambda > 0$ is the regularization parameter and $\mathbf{y}_i^{\uparrow} \in \mathbb{R}^{N_B}$ is the column vector of $\mathbf{Y}^{\uparrow} \in \mathbb{R}^{N_B \times N_H N_W}$ with $\mathbf{Y}^{\uparrow}$ being a bilinear interpolated version of $\mathbf{Y}$.

In order to construct the encoder network, it is unnecessary to solve $\mathbf{x}_i$ and then design $f(\cdot)$, which will lead to error accumulation. We can treat $\mathbf{x}_i$ as an implicit variable and design $f(\cdot)$ by using $\mathbf{z}_i$ and $\mathbf{y}_i^{\uparrow}$ directly, i.e.,

$$\left(\mathbf{z}_i, \mathbf{y}_i^{\uparrow}\right) \xrightarrow{f(\cdot)} \mathbf{s}_i \xrightarrow{g(\cdot)} \hat{\mathbf{x}}_i \quad \forall i. \tag{8}$$

Specifically, we want to obtain the hidden layer output $\mathbf{s}_i$ by solving the following pixelwise fusion model:

$$\min_{\mathbf{s}_i} \frac{1}{2}\|\mathbf{z}_i - \mathbf{RAs}_i\|_2^2 + \frac{\lambda}{2}\left\|\mathbf{y}_i^{\uparrow} - \mathbf{As}_i\right\|_2^2 \quad \forall i \tag{9}$$

and design $f(\cdot)$ by unfolding all steps of its algorithm as network layers. Notably, in (9), both $\mathbf{R}$ and $\mathbf{A}$ are treated as new trainable parameters to facilitate the design of the encoder network. In the model-based HSI super-resolution, our pervious works [17], [51] have shown the effectiveness of (9).

Although (9) has an analytic solution, it is not suitable as the encoder network and is difficult to implement by a network. Equation (9) can be solved by the gradient descent algorithm as

$$\mathbf{s}_i^k = \mathbf{s}_i^{k-1} - \eta\left(\bar{\mathbf{A}}^T\bar{\mathbf{A}}\mathbf{s}_i^{k-1} - \bar{\mathbf{A}}^T\mathbf{z}_i + \lambda\mathbf{A}^T\mathbf{As}_i^{k-1} - \lambda\mathbf{A}^T\mathbf{y}_i^{\uparrow}\right) \tag{10}$$

where $\bar{\mathbf{A}} = \mathbf{RA} \in \mathbb{R}^{N_b \times J}$, $\eta > 0$ is the step, and $k = 1, 2, \ldots, K$ represents the kth iteration. To better design the network, input data and intermediate variables are distinguished by rewriting (10) as

$$\mathbf{s}_i^k = (\mathbf{I} - \eta\bar{\mathbf{A}}^T\bar{\mathbf{A}} - \eta\lambda\mathbf{A}^T\mathbf{A})\mathbf{s}_i^{k-1} + \eta\bar{\mathbf{A}}^T\mathbf{z}_i + \eta\lambda\mathbf{A}^T\mathbf{y}_i^{\uparrow} \tag{11}$$

where $\mathbf{I}$ represents the identity matrix. According to the K iterations of (11), the proposed encoder network is mainly

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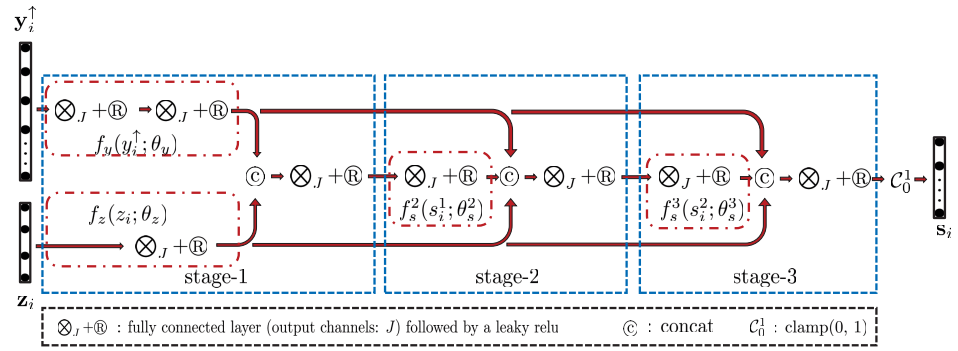


Fig. 2. Details of the proposed encoder network when $K = 3$.

a structure of K stages. Fig. 2 shows the details of $f(\cdot)$ when $K = 3$. In (11), all variables $\mathbf{s}_i^{k-1}$, $\mathbf{z}_i$, and $\mathbf{y}_i^{\uparrow}$ are left multiplied by a matrix. This process is implemented using a fully connected layer followed by a leaky ReLU, and (11) can be rewritten as

$$\mathbf{s}_i^k = f_s^k\left(\mathbf{s}_i^{k-1}; \theta_s^k\right) + \eta f_z(\mathbf{z}_i; \theta_z) + \eta\lambda f_y\left(\mathbf{y}_i^{\uparrow}; \theta_y\right) \tag{12}$$

where $\{f_s^k\}_{k=2}^K$, f_z and f_y, represent the modules designed for the multiplication of matrix and vector and $\{\theta_s^k\}_{k=2}^K$, θ_z and θ_y, represent the trainable parameters involved in the corresponding networks. The red dotted boxes in Fig. 2 show the layers of $\{f_s^k\}_{k=2}^K$, f_z and f_y, where an additional fully connected layer is used in f_y for the purpose of feature extraction. In (12), the three modules are combined linearly. To break the fixed format of optimization model and provide more flexibility, the linear combination is implemented by concatenating these modules and performing a fully connection and a leaky ReLU. Equation (12) can be rewritten as

$$\mathbf{s}_i^k = f^k\left(f_s^k\left(\mathbf{s}_i^{k-1}; \theta_s^k\right), f_z(\mathbf{z}_i; \theta_z), f_y\left(\mathbf{y}_i^{\uparrow}; \theta_y\right); \theta^k\right) \tag{13}$$

where $\{f^k\}_{k=1}^K$ and $\{\theta^k\}_{k=1}^K$ represent the modules and trainable parameters for the linear combination, respectively. Finally, by performing f^k from 1 to K, we can obtain the hidden output by $\mathbf{s}_i = \mathcal{C}_0^1(\mathbf{s}_i^K)$ and have the trainable parameters $\theta = \{\theta_z, \theta_y, \{\theta_s^k\}_{k=2}^K, \{\theta^k\}_{k=1}^K\}$.

C. Loss Function and Training Strategy

To train the autoencoder network (8), we cannot use the loss function (4) anymore since the input data $\mathbf{x}_i$ are just an implicit variable that does not exist. Instead, we have two observed images $\mathbf{Y}$ and $\mathbf{Z}$ to work with and can only build loss function by using the observations and the target $\hat{\mathbf{X}}$. Similar to (5) and (6), the two degenerated images of $\hat{\mathbf{X}}$ are considered, and the outputs LR-HSI $\hat{\mathbf{Y}}$ and HR-MSI $\hat{\mathbf{Z}}$ can be modeled as $\hat{\mathbf{Y}} \approx \hat{\mathbf{X}}\mathbf{BD}$ and $\hat{\mathbf{Z}} \approx \mathbf{R}\hat{\mathbf{X}}$. Then, the loss function (4) can be rewritten as

$$\mathcal{L} = \|\mathbf{Z} - \hat{\mathbf{Z}}\|_{1,1} + \gamma \|\mathbf{Y} - \hat{\mathbf{Y}}\|_{1,1} \tag{14}$$

where $\gamma > 0$ ($\gamma = 1$ is used in this work) is used to balance the spatial and spectral terms.

When training the network (8) using the loss function (14), one has to combine all pixels $\hat{\mathbf{x}}_i$ into an image $\hat{\mathbf{X}}$ since $\hat{\mathbf{Y}}$ and $\hat{\mathbf{X}}$ are coupled together by $\mathbf{BD}$. In other words, one has to train (8) by feeding $\mathbf{Z}$ and $\mathbf{Y}^{\uparrow}$ entirely. Despite this, we can still train (8) by using small patches to accelerate the training process. Specifically, we divide $\mathbf{Z}$ and $\mathbf{Y}^{\uparrow}$ into overlapped patches so that the patches cover all pixels and then discard the pixels affected by spatial blur $\mathbf{B}$ at the boundaries of these patches when computing the loss function (14).

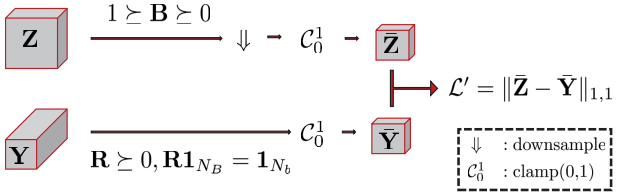


Fig. 3. Details of the blind estimation network.

D. Blind Estimation Network

The PSF $\mathbf{B}$ and the SRF $\mathbf{R}$ are required to train the proposed autoencoder network. By combining (5) and (6), we have

$$\mathbf{ZBD} \approx \mathbf{RY}. \tag{15}$$

By imposing some physical constraints, one can obtain $\mathbf{B}$ and $\mathbf{R}$ by solving

$$\begin{aligned} &\min_{\mathbf{B},\mathbf{R}} \ \|\mathbf{ZBD} - \mathbf{RY}\|_{1,1} \\ &\text{s.t. } 1 \succeq \mathbf{B} \succeq 0, \quad \mathbf{R} \succeq 0, \ \mathbf{R}\mathbf{1}_{N_B} = \mathbf{1}_{N_b}. \end{aligned} \tag{16}$$

Problem (16) can be solved by some optimization algorithms. Instead, we solve (16) by training a network. Specifically, we treat $(\mathbf{Z}, \mathbf{Y})$ as inputs and $(\mathbf{B}, \mathbf{R})$ as trainable parameters. Then, the blind estimation network can be constructed by using the following loss function:

$$\mathcal{L}' = \|\bar{\mathbf{Z}} - \bar{\mathbf{Y}}\|_{1,1} \tag{17}$$

where $\bar{\mathbf{Z}} = \mathcal{C}_0^1(\mathbf{ZBD})$ and $\bar{\mathbf{Y}} = \mathcal{C}_0^1(\mathbf{RY})$ represent the output data. The details of the blind estimation network are shown in Fig. 3. To implement the physical constraints, the trainable parameters $(\mathbf{B}, \mathbf{R})$ are projected into the corresponding ranges every iteration when training the network.

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E. Relationship to Model-Based Approaches

The proposed MIAE can be regarded as a kind of specific fusion model. By combining (1), (8), and (14), MIAE can be rewritten as

$$\begin{aligned} &\min_{\mathbf{A},\theta} \ \|\mathbf{Z}-\mathbf{RAS}\|_{1,1}+\gamma\|\mathbf{Y}-\mathbf{ASBD}\|_{1,1} \\ &\text{s.t. } \mathbf{s}_i = f\left(\mathbf{z}_i,\mathbf{y}_i^{\uparrow};\theta\right) \quad \forall i \\ &\qquad 1\succeq \mathbf{A}\succeq 0, \quad 1\succeq \mathbf{S}\succeq 0, \ 1\succeq \mathbf{AS}\succeq 0. \end{aligned} \tag{18}$$

In (18), $f(\cdot)$ can be thought of as a nonlinear mapping function, and each $\mathbf{s}_i$ is only associated with the input $\mathbf{z}_i$ and $\mathbf{y}_i^{\uparrow}$ that correspond to its spatial position. The constraint $\mathbf{s}_i = f(\mathbf{z}_i,\mathbf{y}_i^{\uparrow};\theta)$ acts as a manifold regularization that embeds the combination of $\mathbf{z}_i$ and $\mathbf{y}_i^{\uparrow}$ into a low-dimensional space $\mathbb{R}^J$. $\mathbf{s}_i = f(\mathbf{z}_i,\mathbf{y}_i^{\uparrow};\theta)$ also acts as a self-supervised deep prior regularization that only uses itself as training data.

III. Experimental Results and Analysis

In this section, experiments on both synthetic and real datasets are conducted to demonstrate the performance of the proposed MIAE. Before the following experiments, all datasets are scaled to the range [0, 1]. The quality of the fused images in the synthetic datasets is assessed with root-mean-squared error (RMSE), peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM), relative dimensionless global error in synthesis (ERGAS), and universal image quality index (UIQI) [2], [3], [5].

A. Synthetic Datasets and Implementation Details

Three real-life HSI datasets, University of Pavia (PaviaU), Kennedy Space Center (KSC), and Washington DC Mall (DC), are manipulated to use as synthetic reference images for the simulation experiments.

1) The PaviaU dataset is acquired by the Reflective Optics System Imaging Spectrometer (ROSIS), with a spectral range of 0.43–0.86 μm. The ROSIS sensor is characterized by 115 spectral bands and 103 remained after removal of noisy bands. This image, with a size of 610×340 pixels, has a spatial resolution of 1.3 m per pixel. We select the up-left 512×256 pixel part as the reference image.
2) The KSC dataset is acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), with a spectral range of 0.4–2.5 μm. The AVIRIS sensor is characterized by 224 spectral bands and the number of spectral bands is reduced to 176 by removing water absorption bands. The size of this image is 512×614 with a spatial resolution of 18 m. We select the up-left 512×512 pixel part as the reference image.
3) The DC dataset is acquired by the hyperspectral digital imagery collection experiment (HYDICE) image, with a spectral range of 0.4–2.4 μm. The HYDICE sensor is characterized by 210 spectral bands, and bands in the region where the atmosphere is opaque have been removed, leaving 191 bands. This image, with a size of 1208×307 pixels, has a spatial resolution of about 2.8 m. We select a 512×256 pixel part as the reference image.

TABLE I
Quality Measures of Nonblind and Blind MIAE

		PaviaU		KSC		DC	
	best	nonblind	blind	nonblind	blind	nonblind	blind
RMSE	0	0.0169	0.0172	0.0426	0.0427	0.0127	0.0145
PSNR	$+\infty$	37.57	37.33	34.21	34.05	37.48	35.90
SAM	0	2.41	2.43	6.98	7.00	1.56	1.84
ERGAS	0	0.647	0.656	3.122	3.129	14.184	14.224
UIQI	1	0.988	0.988	0.882	0.882	0.983	0.975

For each reference image, we generate the two observed images, LR-HSI and HR-MSI, according to the Walds protocol [67]. To generate the LR-HSI, we spatially blur the reference image and then downsample it by a factor of 8 ($r = 8$) in each direction. A Gaussian blur of 15×15 pixels, with a mean of 0 and a standard deviation of 3.40, is applied to each band of the reference image. To generate the HR-MSI, $\mathbf{R}$ is derived from the spectral response of the IKONOS satellite. We generate a four-band image by averaging the bands of the reference image according to the spectral response profiles of the RGB and NIR bands. To account for ubiquitous noise or error, moderate Gaussian noise is added to the LR-HSI (SNR = 30 dB) and the HR-MSI (SNR = 40 dB).

We implement and train the proposed network and blind estimation network using the PyTorch framework. As discussed in Section II-C, we divide the observed images into patches to accelerate the training process. Taking the HR-MSI as a reference, 40×40-pixel overlapping patches with a stride of 24 are extracted for training. The batch sizes are 25 for the PaviaU and DC datasets and 50 for the KSC dataset. An Adam optimizer is used to train the network for 10 000 iterations. The learning rate is initialized as 5×10^{-3} and gradually decayed by multiplying $1 - (1/9000)\max(0, \text{iteration} - 1000)$, where 'iteration' represents the current number of iterations. As for the blind estimation network, it is trained by feeding the observed images entirely, the total number of iterations is 5000, and the learning rate is set as 5×10^{-5}. The code of the proposed MIAE will be available on https://github.com/liuofficial.

B. Influence of Parameters

Two parameters, rank J and stage K, need to be given when constructing the proposed network. In this set of experiments, we investigate them and show how they impact the quality measures of MIAE. Fig. 4 shows the PSNR results of MIAE as a function of J when $K = 1, 2, \ldots, 5$. It can be seen that, for all datasets, the PSNR performance improves as J increases, but a large J will cause overfitting or performance degradation. Compared with small values of K, a moderate K is better and a too-large K is prone to overfitting. Thus, K is eventually set as 3 for all datasets, and J is eventually set as 80 for the PaviaU and KSC datasets and 30 for the DC dataset.

C. Experiment Results on Synthetic Datasets

1) Blind and Nonblind: Section II-D presents a blind estimation network for estimating the PSF and SRF.

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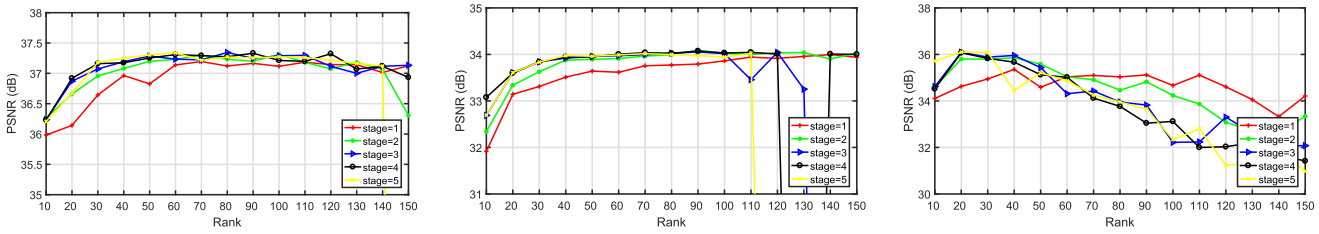


Fig. 4. PSNR as a function of rank J when using different stages K. (a) PaviaU dataset. (b) KSC dataset. (c) DC dataset.

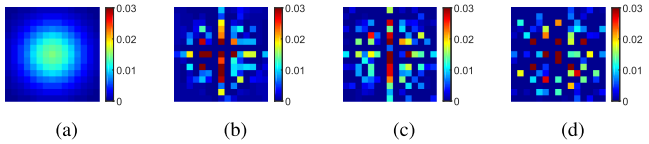


Fig. 5. Illustration of the true and estimated PSF. (a) True. (b) PaviaU dataset. (c) KSC dataset. (d) DC dataset.

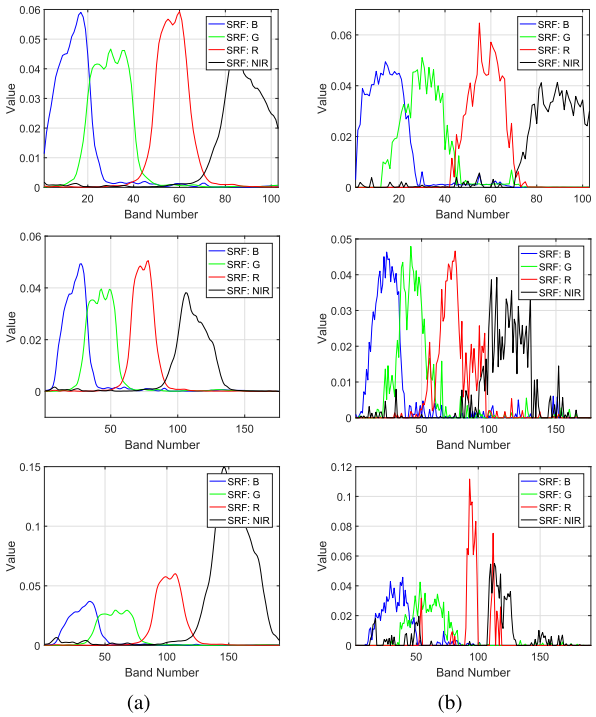


Fig. 6. Illustration of the true and estimated SRF. (Top) PaivaU dataset. (Middle) KSC dataset. (Bottom) DC dataset. (a) True SRF. (b) Estimated SRF.

This experiment is used to evaluate the estimated **B** and **R**. Figs. 5 and 6 show the estimated **B** and **R**, respectively, where the exact ones are also included for reference. The estimated **B** exhibits a sparse property and the curves of the estimated **R** are relatively coarse. Table I shows the quality measures of the proposed MIAE using the exact and estimated **B** and **R**, that is, nonblind and blind cases. It can be seen that, for the PaviaU and KSC datasets, the performance degradation caused by blind estimation is very small when compared with the nonblind estimation, and for the DC dataset, the performance degradation is also not significant.

2) Influence of LR-HSI Interpolation: For the proposed MIAE, bilinear interpolation is used to upsample the LR-HSI to the same size of the target HR-HSI. This experiment shows how the interpolation method affects the performance of MIAE. Four interpolation methods are considered, i.e., bilinear interpolation, nearest interpolation, bicubic interpolation, and cubic spline interpolation. The quality measures to assess the different interpolation methods are given in Table II. In most cases, there is no obvious difference between these interpolation methods. The nearest interpolation performs slightly worse on the KSC dataset and the bicubic interpolation on the DC dataset.

3) Comparison With the State of the Art: Nine unsupervised methods, which can be divided into model- and deep learning-based approaches, are compared to evaluate the performance of MIAE. The model-based approaches consist of six methods. The first method is the baseline one (denoted by SYLV[1]) that solves a Sylvester equation [68], and the next five methods are coupled NMF (CNMF[2]) [18], CSU[3] [19], NSSR[4] [22], HySure[5] [13], and NPTSR[6] [11]. The deep learning-based approaches are CNNFUS[7] [56] and three autoencoder-based methods, i.e., uSDN[8] [58], HyCoNet[9] [62], and also MIAE. The free parameters of the compared methods are tuned to be optimal with the test datasets, and the default training strategies are used for the deep learning-based approaches. All of the compared methods are blind, where the estimated **B** and **R** are used. For those methods that do not involve blind estimation procedures, **B** and **R** are estimated by the proposed blind estimation network.

The five quantitative results of the compared methods for the PaviaU dataset are shown in Table III with the best values marked in bold. It can be seen that all the methods outperform the baseline method SYLV, and the proposed MIAE method gives the best quantitative results followed by NPTSR and HyCoNet. Both the model- and deep learning-based approaches can yield good results. Fig. 7 shows the reference image and the fusion results of the compared methods in form of the 30th band gray and error images. Visually, it can be observed that the baseline method SYLV has severe spatial

[1]The code is derived from the toolbox on the website: https://openremotesensing.net/wp-content/uploads/2017/11/HSMSFusion Toolbox.zip

[2]https://naotoyokoya.com/assets/zip/CNMF_MATLAB.zip

[3]https://github.com/lanha/SupResPALM

[4]http://see.xidian.edu.cn/faculty/wsdong/wsdong_Publication.htm

[5]https://github.com/alfaiate/HySure

[6]The code is provided by Dr. Xu.

[7]https://github.com/renweidian/CNN-FUS

[8]https://github.com/aicip/uSDN

[9]https://github.com/saber-zero/HyperFusion

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TABLE II

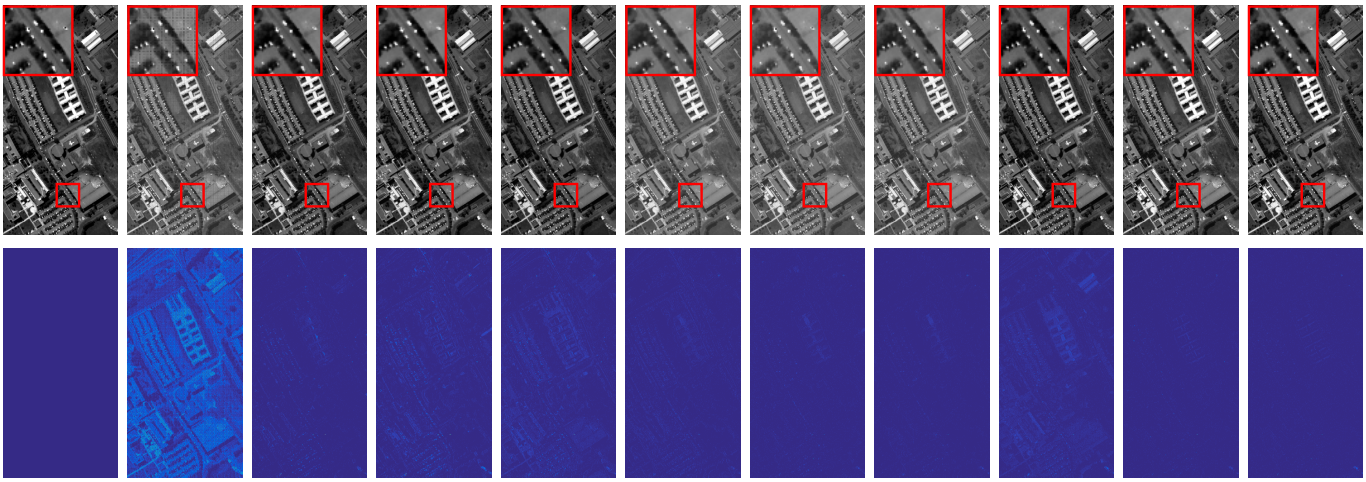
QUALITY MEASURES OF MIAE USING DIFFERENT INTERPOLATION METHODS

		PaviaU				KSC				DC			
	best	bilinear	nearest	bicubic	spline	bilinear	nearest	bicubic	spline	bilinear	nearest	bicubic	spline
RMSE	0	0.0172	0.0172	0.0171	0.0172	0.0427	0.0431	0.0428	0.0426	0.0145	0.0151	0.0148	0.0142
PSNR	$+\infty$	37.33	37.28	37.30	37.27	34.05	33.88	34.02	34.04	35.90	35.61	34.77	35.78
SAM	0	2.43	2.43	2.41	2.43	7.00	7.11	7.01	7.01	1.84	1.93	1.83	1.82
ERGAS	0	0.656	0.656	0.652	0.653	3.129	3.164	3.135	3.130	14.224	14.204	14.349	14.188
UIQI	1	0.988	0.988	0.988	0.988	0.882	0.879	0.882	0.881	0.975	0.974	0.975	0.975

TABLE III

QUALITY MEASURES FOR THE PAVIAU DATASET USING DIFFERENT METHODS (THE BEST VALUES ARE HIGHLIGHTED)

Method	SYLV	CNMF	CSU	NSSR	HySure	NPTSR	CNNFUS	uSDN	HyCoNet	MIAE
RMSE	0.1072	0.0196	0.0231	0.0236	0.0194	0.0186	0.0237	0.0258	0.0188	**0.0172**
PSNR	23.79	35.75	33.87	33.93	36.09	36.71	35.24	32.84	36.67	**37.33**
SAM	12.62	2.62	2.89	3.21	2.70	2.64	3.16	3.49	2.66	**2.43**
ERGAS	3.646	0.741	0.849	0.871	0.728	0.699	0.825	0.905	0.720	**0.656**
UIQI	0.853	0.987	0.983	0.983	0.986	0.987	0.984	0.982	0.987	**0.988**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 7. Images (with a meaningful region marked and zoomed in three times for easy observation) and error maps at band 30 of HSI super-resolution results when applied to the PaviaU dataset. (a) Reference image. (b) SYLV. (c) CNMF. (d) CSU. (e) NSSR. (f) HySure. (g) NPTSR. (h) CNNFUS. (i) uSDN. (j) HyCoNet. (k) MIAE.

TABLE IV

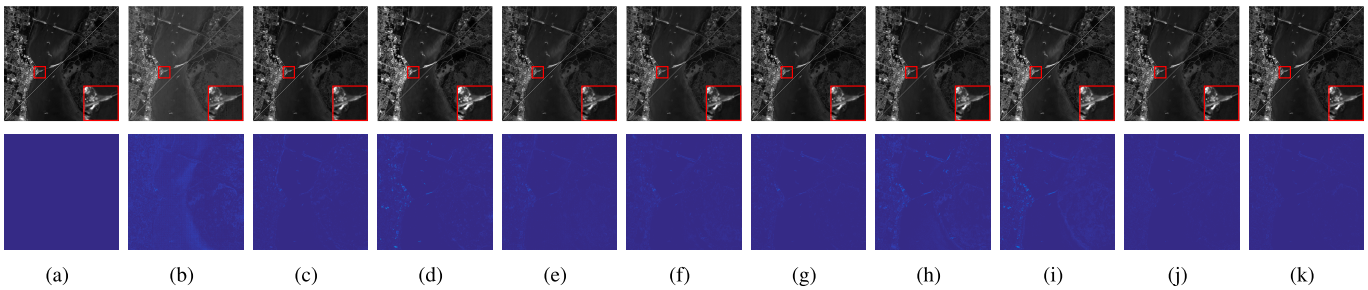
QUALITY MEASURES FOR THE KSC DATASET USING DIFFERENT METHODS (THE BEST VALUES ARE HIGHLIGHTED)

Method	SYLV	CNMF	CSU	NSSR	HySure	NPTSR	CNNFUS	uSDN	HyCoNet	MIAE
RMSE	0.1574	0.0454	0.0465	0.0513	0.0453	0.0450	0.0534	0.0504	0.0441	**0.0427**
PSNR	19.33	32.70	31.17	30.77	32.95	33.30	30.95	30.06	33.49	**34.05**
SAM	23.23	7.78	8.02	8.65	7.64	7.29	9.01	9.14	7.22	**7.00**
ERGAS	8.753	3.497	3.405	3.738	3.336	3.328	3.954	3.717	3.262	**3.129**
UIQI	0.506	0.870	0.855	0.836	0.881	**0.887**	0.843	0.857	0.878	0.882

distortion and all other methods outperform it. MIAE and HyCoNet perform better than the other methods in terms of both zoomed region and error map. Fig. 10(a) shows the PSNR as a function of spectral band for the compared methods. It can be seen that the proposed MIAE method performs best in almost all bands followed by HyCoNet and NPTSR. Fig. 11(a) shows the SAM between the reference image and the fusion results for each pixel using the compared methods, with the pixels sorted by ascending error. As illustrated in this figure, MIAE consistently outperforms the others at the pixel level.

Table IV reports the five quality measures of the compared methods for the KSC dataset. From this table, we can see that the baseline method SYLV performs the worst, NPTSR gives the best UIQI result, and the proposed MIAE method performs best for the remaining four quality measures. NPTSR and HyCoNet are only inferior to MIAE. In Fig. 8, we show the reference image and the fusion results of the compared methods in form of the 30th band gray and error images. Visually, it can be observed that the reconstructed results of HyCoNet and MIAE are better than the others, and the baseline method SYLV gives the worst images. Fig. 10(b) shows PSNR as a function of the spectral band for the compared methods. MIAE, HyCoNet, and NPTSR achieve high results in most bands. Fig. 11(b) shows the SAMs for

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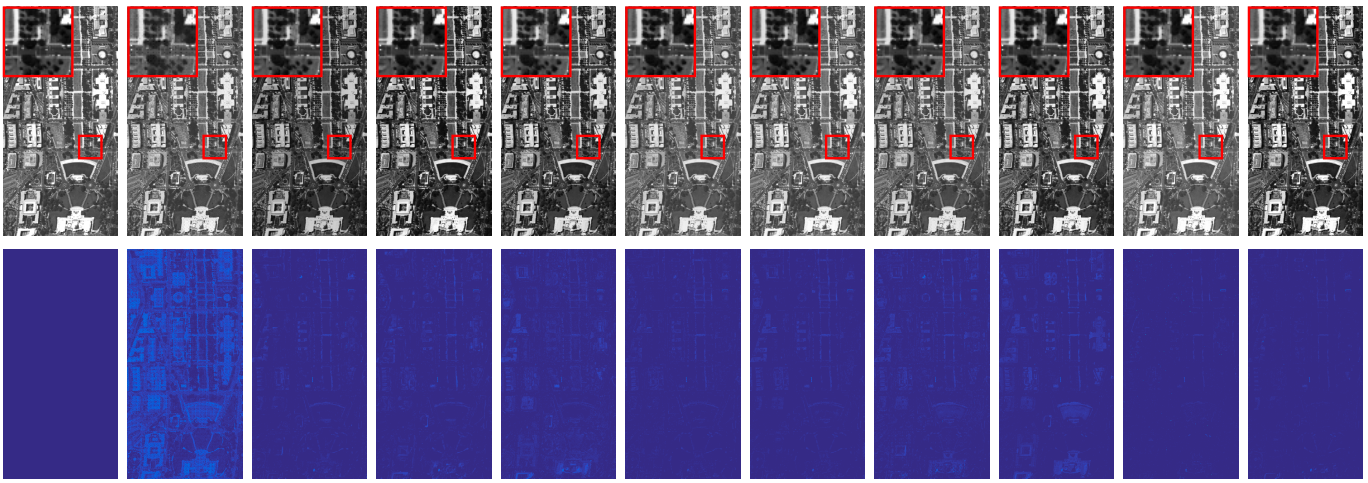


(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 8. Images (with a meaningful region marked and zoomed in three times for easy observation) and error maps at band 30 of HSI super-resolution results when applied to the KSC dataset. (a) Reference image. (b) SYLV. (c) CNMF. (d) CSU. (e) NSSR. (f) HySure. (g) NPTSR. (h) CNNFUS. (i) uSDN. (j) HyCoNet. (k) MIAE.

TABLE V
QUALITY MEASURES FOR THE DC DATASET USING DIFFERENT METHODS (THE BEST VALUES ARE HIGHLIGHTED)

Method	SYLV	CNMF	CSU	NSSR	HySure	NPTSR	CNNFUS	uSDN	HyCoNet	MIAE
RMSE	0.1685	0.0283	0.0220	0.0366	0.0275	0.0261	0.0332	0.0278	0.0197	**0.0145**
PSNR	18.53	32.61	32.48	29.90	32.15	32.77	32.39	29.24	31.63	**35.90**
SAM	26.56	3.82	2.82	5.33	3.48	3.54	3.70	3.53	**1.83**	1.84
ERGAS	37.085	15.182	14.406	14.447	13.797	13.886	14.300	17.345	**10.943**	14.224
UIQI	0.469	0.930	0.929	0.905	0.930	0.944	0.954	0.856	0.914	**0.975**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 9. Images (with a meaningful region marked and zoomed in three times for easy observation) and error maps at band 30 of HSI super-resolution results when applied to the DC dataset. (a) Reference image. (b) SYLV. (c) CNMF. (d) CSU. (e) NSSR. (f) HySure. (g) NPTSR. (h) CNNFUS. (i) uSDN. (j) HyCoNet. (k) MIAE.

each pixel between the reference image and the fusion results, with the pixels sorted in order of ascending error. It can be observed that MIAE is the best followed by HyCoNet and NPTSR.

Table V summarizes the five quality measures of the compared methods for the DC dataset. From this table, we can see that MIAE gives three best quantitative results and one second best, and HyCoNet gives two best. The 30th band gray and error images of the reference image and the fusion results of the compared methods are given in Fig. 9. Through visual inspection, we can see that MIAE and HyCoNet exhibit good reconstructed results. PSNR and SAM, as functions of the spectral band and by pixel sorted on error, are shown in Figs. 10(c) and 11(c), respectively. It can be seen that MIAE outperforms the others in terms of band-level PSNR, and MIAE and HyCoNet achieve higher results than the others in terms of pixel-level SAM.

D. Experiment Results on Real Data

The University of Houston (UH) dataset released by the 2018 IEEE GRSS Data Fusion Contest [69] is used to evaluate MIAE in practical applications. The original data are acquired by the National Center for Airborne Laser Mapping (NCALM), covering the UH campus and its surrounding urban areas. This experiment selects an LR-HSI and a high-resolution RGB (HR-RGB) image from this multimodal optical remote sensing datasets. The LR-HSI collected by the ITRES CASI-1500 sensor contains 4172×1202 pixels with a spatial resolution of 1 m and 48 spectral bands with a spectral range of 0.38–1.05 μm. The HR-RGB image collected by DiMAC ULTRALIGHT+ sensor contains $83\,440 \times 24\,040$ pixels. Taking the LR-HSI as a reference, we select an area of $64 \times 64 \times 48$ as our observation data and downsample the corresponding area of the HR-RGB to be a $512 \times 512 \times 3$ size

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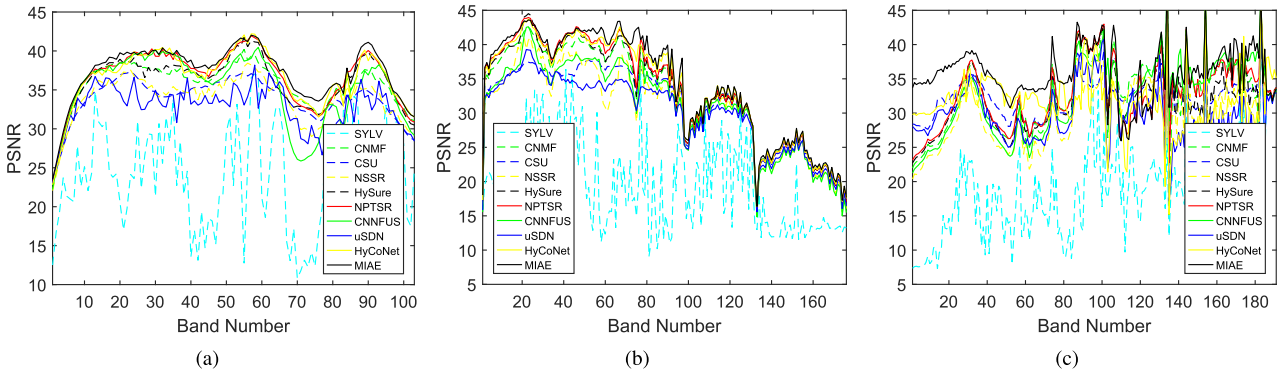


Fig. 10. PSNR as a function of spectral band. (a) PaviaU dataset. (b) KSC dataset. (c) DC dataset.

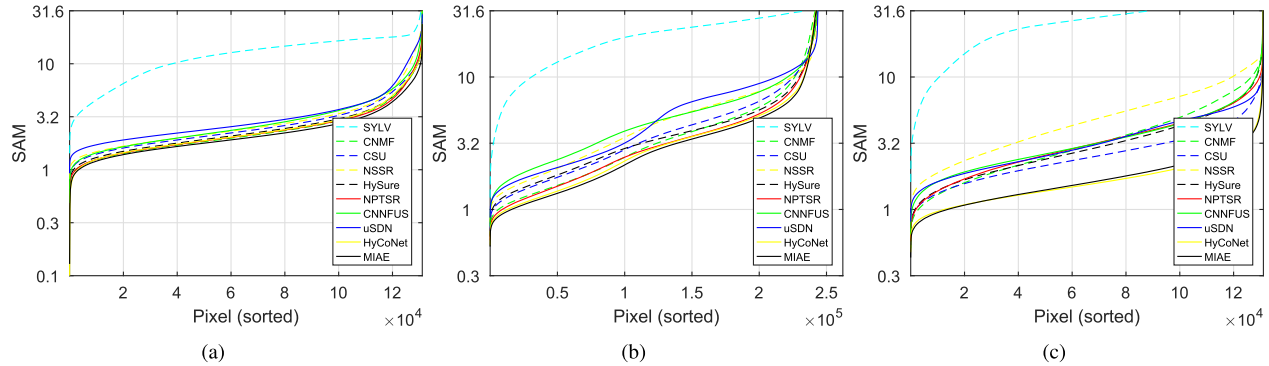


Fig. 11. SAM (plotted in a $\log_{10}(\cdot)$ scale) as a function of sorted pixel. (a) PaviaU dataset. (b) KSC dataset. (c) DC dataset.

TABLE VI

QUALITY MEASURES WITHOUT REFERENCE FOR THE UH DATASET USING DIFFERENT METHODS (THE BEST VALUES ARE HIGHLIGHTED)

	Best	SYLV	CNMF	CSU	NSSR	HySure	NPTSR	CNNFUS	uSDN	HyCoNet	MIAE
D_λ	0	0.1883	0.1687	0.0575	0.1548	0.0683	0.1391	0.1321	**0.0523**	0.0708	0.0898
D_s	0	0.1248	0.2735	0.1814	0.2786	0.1938	0.2405	**0.0711**	0.1456	0.1286	0.0987
QNR	1	0.7104	0.6039	0.7715	0.6097	0.7511	0.6539	0.8062	0.8097	0.8097	**0.8204**

TABLE VII

RUNNING/TRAINING TIMES (IN SECONDS) OF THE COMPARED METHODS

Method	SYLV	CNMF	CSU	NSSR	HySure	NPTSR	CNNFUS	uSDN	HyCoNet	MIAE-ONE	MIAE
PaviaU	1.6	16.4	156.9	181.9	167.6	1339.0	9.4	504.5	562.6	541.4	186.5
KSC	5.6	36.5	302.7	499.0	336.1	4599.2	14.1	827.9	970.7	959.1	396.4
DC	3.0	16.5	172.2	290.6	171.1	2424.3	8.8	477.8	577.9	525.0	239.1
UH	2.0	19.1	47.1	109.4	169.9	1262.1	9.0	255.8	726.5	449.9	155.9

image, that is, the resolution ratio is $r = 8$. RGB images of the real dataset and the fusion results of the compared methods mentioned in Section III-C3 are given in Fig. 12. Visually, it can be seen that MIAE, NPTSR, and HySure give the good color and brightness results, and the result of the proposed MIAE is much closer to the HR-RGB image. Since the reference HR-HSI is not available, three quality measures without reference, spectral distortion index D_λ, spatial distortion index D_s, and quality with no reference (QNR) [6], [70], are adopted to quantitatively measure the quality of the fusion results. For D_s, it is the average of the evaluation results of the three RGB channels. Table VI shows the three quality measures of the compared methods. It can be seen that the proposed MIAE gives the highest QNR.

E. Computational Efficiency

All of the above experiments are carried out using a desktop computer with an Intel Core i9-7900X CPU, a GeForce GTX 2080Ti GPU, and 64-GB memory. The computational efficiency of the compared methods mentioned in Section III-C3 is investigated. The first seven methods, SYLV, CNMF, CSU, NSSR, HySure, NPTSR, and CNNFUS, are performed using MATLAB, and the remaining three autoencoder-based methods, uSDN, HyCoNet, and MIAE, are implemented by the PyTorch framework. Table VII summarizes the running times of the first seven methods and the training times of the autoencoder-based methods, and the number of trainable parameters for each autoencoder network is reported in Table VIII.

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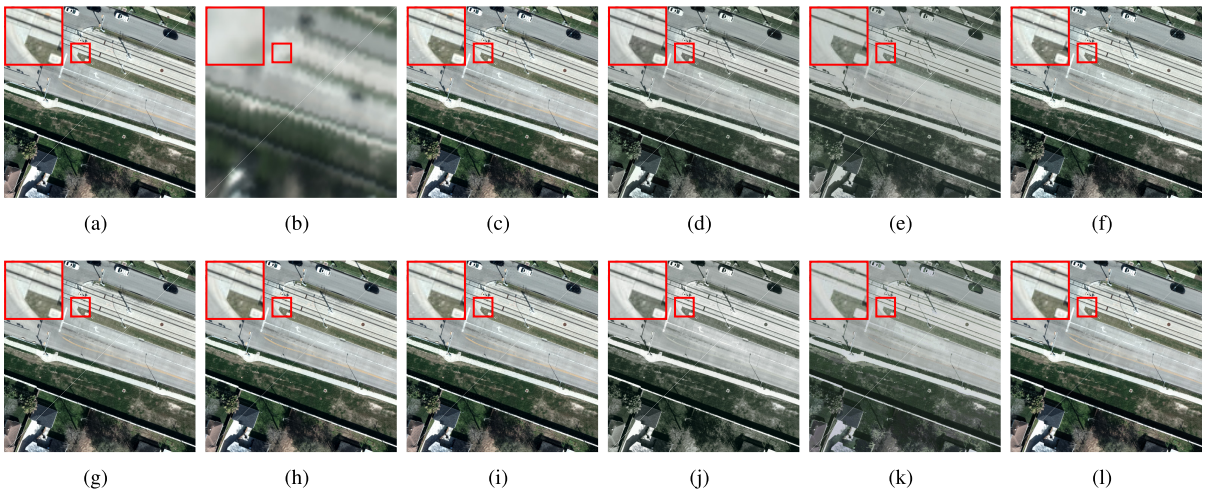


Fig. 12. RGB images (with a meaningful region marked and zoomed in three times for easy observation) of HSI super-resolution results when applied to real dataset. (a) HR-RGB image. (b) LR-HSI. (c) SYLV. (d) CNMF. (e) CSU. (f) NSSR. (g) HySure. (h) NPTSR. (i) CNNFUS. (j) uSDN. (k) HyCoNet. (l) MIAE.

TABLE VIII
NUMBER OF TRAINABLE PARAMETERS

Method	uSDN	HyCoNet	MIAE
PaviaU	37.9K	377.7K	87.8K
KSC	48.9K	389.5K	99.5K
DC	51.1K	391.9K	21.7K
UH	2.3K	368.6K	13.1K

For the proposed MIAE, the training times of feeding the observed images entirely are also included in Table VII for reference, and the corresponding method is denoted by MIAE-ONE. It can be seen that MIAE takes less time to train the network than the other two autoencoder-based methods, and its trainable parameters are much less than HyCoNet. Ignoring the platform, MIAE is comparable to the model-based approaches.

IV. CONCLUSION

This article has proposed an unsupervised MIAE network for HSI super-resolution. The proposed MIAE involves an implicit autoencoder network and the structures are concise. First, inspired by that performing NMF on the target HR-HSI can facilitate the inference process of super-resolution, the implicit autoencoder network is built on the target HR-HSI by integrating its NMF model, where the two NMF parts, spectral and spatial matrices, are treated as decoder parameters and hidden outputs respectively. The autoencoder network treats each hyperspectral pixel of the target HR-HSI as an individual sample, that is, the network is trained pixel by pixel. Second, the "implicit" indicates that the input pixel of the autoencoder network is unknown, and thus, a pixelwise fusion model taken the two observed images as inputs is presented to estimate the hidden layer vector directly. The pixelwise fusion model is simple and effective. Specifically, the LR-HSI is resized to the same size of the target HR-HSI using bilinear interpolation, in order to feed the network pixel by pixel. To break the fixed format of model and provide more flexibility, the gradient descent algorithm is used to solve the pixelwise fusion model, and the algorithm is reformulated and unfolded to form the encoder network. Finally, the loss function is built on the relationship between the target HR-HSI and the two observed images. With the specific pixelwise architecture, MIAE can be treated as a kind of manifold prior-based model and can be trained patch by patch to accelerate the training process. Moreover, a blind estimation network is proposed to estimate the PSF and SRF in an unsupervised manner. MIAE has been experimentally tested using three synthetic datasets and one real dataset, and the experimental results demonstrate its effectiveness. Although the results obtained by MIAE are very encouraging, further improvements, such as the application of convolutional autoencoder, should be pursued in future.

ACKNOWLEDGMENT

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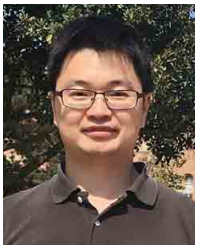
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